Aslam Shaik

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Dynamic Software Engineer with 4 years of experience building scalable backend systems, microservices, and ML-ready data pipelines handling terabyte-scale workloads. Known for architecting resilient platforms in healthcare finance, blending backend engineering with data-driven innovation.

Education

Purdue University DECEMBER 2025
Master of Science in Computer and Information Science (GPA: 3.8 / 4.00) *Indiana*

- **Coursework:** XAI, DataBases, Machine Learning, Distributed Systems, Computer Networks, Algorithms.
- **Graduate Teaching Assistant:** Mentored 100+ undergraduate students as a TA for **Software Systems** delivering hands-on labs on **C++** fundamentals and **Java OOP**, and guiding full-cycle development of software projects.

Technical Skills

Languages: Java, Python, c++, c#, JavaScript.
Technologies: .NET, AWS, SpringBoot, Node.js, Kafka, Flask, React.js, PyTorch, Airflow, MongoDB, SQL, Linux, Git, Redis.
Concepts: System Design, Microservices Architecture, Scalable Distributed Systems, RESTful APIs, CI/CD, Data Structures Algorithms, ML Pipelines, Big Data, SQL/NoSQL, Caching, Concurrency, Agile Development, and Technical Leadership.

Experience

Indiana University August 2024 – Present
Software Engineer, Research in Big Data and AI *Indianapolis, IN*

- **Lead Research Engineer** responsible for the **architecture and design** of a scalable **precipitation forecasting and evapotranspiration tracking platform**, processing **25+ years of climate and farmland datasets**. Designed and implemented **high-throughput data ingestion pipelines** with **ETL optimization, partitioning, and fault-tolerant architecture** to accelerate ML model training and regression analysis; ensured **robust, low-latency, and scalable workflows** for high-volume simulations supporting **food and water security policymaking**.
- Designed and implemented a **cloud-native application architecture** for an NSF-funded healthcare project, building end-to-end pipelines that ingest **real-time sensor data** into **ML models for disease prediction**. Architected **scalable big data workflows** capable of handling **terabyte-scale datasets**, achieving **99% latency reduction** and **65% faster query performance** through advanced **indexing, partitioning, and caching**.

MSI Express, Inc. May 2025 – August 2025
Software Development and Data Engineering Intern *Chicago, IL*

- Architected and delivered a **full-stack production planning platform** from the ground up after analyzing complex manufacturing workflows; built a scalable **React.js frontend**, **Flask microservices backend**, and **Snowflake data warehouse** integration. Automated end-to-end production planning processes, eliminating manual effort and saving **100+ hours per week**, while optimizing system **speed, scalability, and efficiency** across **10+ manufacturing plants**.
- Built high-throughput ETL pipelines (Airflow, Redshift) with **99.99%** uptime, processing **millions of records** per refresh; implemented fault-tolerant retries, partitioning and backpressure handling to scale ingestion while preserving data consistency for analytics.

CAMP Systems International July 2022 – December 2023
Software Engineer, Machine Learning, Backend

- Led the development of a **real-time ML classification system**, designing a **scalable data pipeline** for model evaluation and mentoring an intern on frontend integration. Automated evaluation workflows improved **deployment efficiency by 60%** while maintaining **99.9% uptime**.
- Designed **data pipelines** for near real-time classification and daily model retraining based on performance trends; integrated **AWS Textract** and **Kafka** to extract key data, pass messages between **.NET** enterprise applications and **Python** pipelines, and persist results in the database.

MTX Group Inc January 2022 – June 2022
Software Engineer, Full Stack

- Engineered a banking CRM Salesforce platform for Wells Fargo Bank, N.A. using **C# .NET** and **Java Spring Boot**, designing microservices to migrate customer data and automate processes with **MySQL** integration, boosting data synchronization efficiency by **35%** with clean, modular code.
- Architected robust backend solutions leveraging **ASP.NET, JPA, and PostgreSQL** for seamless data migration, ensuring consistency with **Git** version control and **leveraged design patterns**; enhanced scalability through **unit testing** and **Docker** deployments for reliable CRM workflows.